

---

# Combining Mechanical and Agentic Specification Inference

---

Anonymous Author(s)

Affiliation

Address

email

## Abstract

1 We describe a specification inference tool that combines mechanical weakest-  
2 precondition (WP) analysis with an agentic coding CLI such as Claude Code.  
3 It is implemented for Move and the Move Prover, but the approach is general  
4 enough to be suitable for any Rust-style language. Inference reduces the boilerplate  
5 of verification which often hit applications in practice: to establish a high-level  
6 property such as a global state invariant, pre- and post-conditions for the supporting  
7 functions typically have to be written for the whole software system, a laborious  
8 and tedious task. A Model Context Protocol (MCP) service exposes the WP  
9 analysis and the verifier to the coding agent. The WP analysis provides a sound,  
10 mechanical baseline for inference; the AI is used precisely where WP is weakest —  
11 for loop invariants and high-level idiomatic specifications such as monotonicity,  
12 conservation, and structural invariants. The verifier serves as the oracle that decides  
13 whether the generated specs are valid, and the agent is equipped to generate  
14 proof hints and to refine the inferred specification until verification succeeds.  
15 We evaluate the approach on 17 targets drawn from a private perpetual-trading  
16 codebase, comparing three inference tactics — one without WP and two with —  
17 over four replicates each. At the stated GLM-5.3 prices, the WP-backed tactics cost  
18 about 28% less and leave fewer contracts incomplete, though the unaided agent  
19 still produces a complete specification in 64 of its 68 sessions.

## 20 1 Introduction

21 A roadblock in program verification adoption is the *cost of specification*. The properties developers  
22 want to establish are often high-level: that a call chain aborts only under stated preconditions, that an  
23 API endpoint satisfies postconditions, that data is well-formed with respect to an invariant, or that a  
24 resource cannot be accessed without required permissions. But to verify those properties developers  
25 have to write specifications for every single function in the system.

26 Modern verifiers automatically abstract from imperative code using Dijkstra’s weakest-precondition  
27 transformation (WP) [9]. The problem is that WP stops at loops. Most loops require *loop invariants*,  
28 which eliminate the cycle: the invariant serves as the induction hypothesis, and the loop is replaced  
29 by an acyclic fragment WP can propagate through [6]. There is no complete technique to synthesize  
30 such invariants, even though there has been a lot of work in this area, going back to Cousot and  
31 Halbwachs [8]. Loop invariants have to be provided manually in realistic applications of program  
32 verification. The invariants are not only needed for the code which is the target of verification, they  
33 are also needed for all dependencies, unless those themselves have fully specified function interfaces.

34 Where the human is in the loop to solve tedious problems, LLMs are there to help. Specification  
35 inference via LLMs has been investigated as early as 2023 [20, 15]. While the success rate in these  
36 early years wasn’t high, LLMs have drastically improved since then. In the 2026 era of frontier  
37 models, it appears quite realistic to use a well-instructed agent to infer specifications, including loop

38 invariants, specifically since the agent receives feedback from the verifier: the program must, by  
39 definition, conform to the inferred specification, and if not, a decent verifier will produce a meaningful  
40 diagnosis.

41 In this work we are going one step further: we supply the agent with a WP tool which can infer  
42 precise, simplified specifications, provided loop invariants are given. The intended workflow is that  
43 the agent requests specs for a program, the tool comes back either with success, then the task is done,  
44 or with information about the missing loop invariants. The agent then asks the model to infer those  
45 invariants, and we loop back to the start.

46 The most interesting question with such a setup is whether the capability to run WP actually improves  
47 specification inference, and whether the overhead introduced by model/tool interaction is worth it.  
48 We investigate this question empirically by comparing pure agent and combined agent/WP on a  
49 corpus of examples drawn out of a private codebase (that of a production perpetual trading system).  
50 The fact that we choose a private codebase is methodologically important, as discussed later. The  
51 currency of comparison is US dollars under the stated token prices; we use GLM-5.3 [23] as the  
52 model and the Claude Code SDK as the harness. The results show that WP-based inference, at least  
53 with today’s models, beats the agent-only approach on average by 28% lower cost.

54 The combination of agentic and mechanical inference as discussed here has been implemented in the  
55 Move Prover (MVP). MVP has been developed as part of the Libra/Diem project at Meta, targeting  
56 the Move smart contract language, and later transitioned to the blockchain company that maintains it.  
57 The Move language has many similarities to Rust: safe reference semantics, linear types, algebraic  
58 data types, higher-order functions. Those traits are what the mechanical half of our approach relies  
59 on — specifically the ability to eliminate references (Sec. 3) — so while we implement and evaluate  
60 it for Move, we expect it to carry over to verifiers for Rust and similar languages. Move has been  
61 used to specify and verify Diem’s international payment system and also for critical components of a  
62 public blockchain [19, 3]. The Prover supports pre/post conditions, abort conditions, state/data/loop  
63 invariants, and higher-order functions, in first-order predicate logic. It has an efficient encoding in  
64 SMT/z3 via Boogie, which has been described in [10, 13]. For more details, see the Move book [4].

## 65 2 Inference Session Example

66 We illustrate the workflow of specification inference in our system by examples. The examples are  
67 written in Move, but should be understandable without knowing details about the language. For a  
68 reference, see the Move Book [4].

69 Consider a coding agent like Claude Code. The Move Prover is connected to the agent via an MCP  
70 service. The service offers (a) a tool for launching verification and (b) a tool for running WP-based  
71 spec inference. The agent has a set of skills which describe language, verification techniques, and the  
72 MCP tools in detail.

73 The example we look at is an iterative computation of the exponential,  $b^e$ :

```
74 fun pow(base: u64, exp: u64): u64 {  
    let result = 1;  
    let i = 0;  
    while (i < exp) {  
        result = result * base;  
        i = i + 1;  
    }  
    result  
}
```

75 Based on the instructions, the first thing the agent does is call WP. The answer points to the missing  
76 loop invariant, including unrolling the loop symbolically and showing the values of the affected loop  
77 variables at each entry:

```
78 WP inferred ‘vacuous’ conditions after this loop without an invariant. . .  
--> sources/pow.move:5:14  
|
```

```

5 |         while (i < exp) {
|           ^
|
| = loop-invariant evidence
| = bounded loop-head facts :
|   head[0]: head[0].i = 0
|             head[0].result = 1
|   head[1]: 0 < exp  $\implies$  head[1].i = 1
|             0 < exp  $\implies$  head[1].result = base
|   head[2]: 1 < exp  $\implies$  head[2].i = 2
|             1 < exp  $\implies$  head[2].result = base * base
|   ..

```

79

80 Based on the diagnosis, the agent next derives the loop invariant. A recursive specification helper  
81 function is introduced for accumulating the result. Before returning the result, the agent uses the  
82 verification tool to check whether its solution verifies:

```

fun pow(base: u64, exp: u64): u64 {
  ..
  while (i < exp) {
    result = result * base;
    i = i + 1;
  } spec {
    invariant [inferred] i  $\leq$  exp;
    invariant [inferred] result = math_pow(base, i);
  }
  ..
}
// specification helper function
spec fun math_pow(base: num, exp: num): num {
  if (exp  $\leq$  0) { 1 } else { math_pow(base, exp - 1) * base }
}

```

agent

83

84 In the next turn the agent runs the WP MCP tool again. With the loop invariant present, the complete  
85 specification is derived. However, the aborts condition is marked as sathard by the WP tool. The  
86 marker is set when a derived condition is exact but its satisfiability is expected to be hard for the solver  
87 — here because of the existential quantification, which says that the computation will abort if for  
88 some y the exponent multiplication overflows. Note that the Move Prover uses unbounded integers,  
89 num, in the specification language, making it possible to reason about overflow in the programming  
90 language's u64:

```

spec pow { // post and aborts condition for pow
  ensures [inferred] result = math_pow(base, exp);
  aborts_if [inferred = sathard]  $\exists$  y: u64:
    math_pow(base, y) * base > MAX_U64
    && ( $\forall$  x: num: 0  $\leq$  x && x  $\leq$  y  $\implies$  math_pow(base, x)  $\leq$  MAX_U64)
    && y < exp
}

```

wp tool

91

92 The agent tries to verify this specification, but it times out with the following diagnosis:

```

verification failed:
error: verification out of resources/timeout (timeout set to 5s)
= timeout analysis replay: mixed quantifier and nonlinear arithmetic
= top quantifier activity:
  quantifier at sources/pow.move:24 -- 55,793+ observed instantiations

```

verify tool

93

94 The quantifier the diagnosis is pointing to is the defining recursive axiom of the specification  
95 helper function math\_pow, which is apparently endlessly unfolded in order to prove the sathard  
96 aborts condition. In the next turn, the agent introduces lemmas to characterize the monotonicity of  
97 multiplication, which allows them to rewrite the aborts condition:

```

spec pow {
  pragma opaque;
  ensures [inferred] result = math_pow(base, exp);
  /// The partial products are non-decreasing, so the loop
  /// overflows exactly when its last multiplication does.
  aborts_if [inferred] exp > 0 && math_pow(base, exp - 1) * base > MAX_U64;
} proof {
  /// Bounds the multiplication of every iteration by the last one; the
  /// trigger fires on the 'math_pow(base, i)' of the loop invariant.
  ∀ k: num {math_pow(base, k)} apply math_pow_last_is_max(base, k, exp);
}
spec lemma math_pow_last_is_max(base: num, k: num, n: num) {
  requires base ≥ 0 && 0 ≤ k && k < n;
  ensures math_pow(base, k) * base ≤ math_pow(base, n - 1) * base;
} proof {
  apply math_pow_mono(base, k + 1, n);
}
spec lemma math_pow_mono ... // left out for brevity

```

98

### 99 3 Weakest-Precondition Analysis

100 We run a textbook WP analysis [9, 6] over the bytecode MVP hands to its Boogie back-end, and read  
 101 off abort conditions, modifies clauses, and consequence-style postconditions. The novelty here is not  
 102 in the WP algorithm, but in the *shape of the bytecode it operates on*: by the time the analysis runs,  
 103 MVP’s standard transformation pipeline has already eliminated all Move references and injected  
 104 every available specification as in-line assume/assert statements. The result is a reference-free,  
 105 contract-annotated control flow on which backward predicate propagation reduces to the classical  
 106 case.

107 **Reference elimination.** Move has Rust-style references:  $\&T$  for shared reads and  $\&\text{mut } T$  for exclusive  
 108 writes, both with a stack-bounded lifetime. Predicate-transformer reasoning over such references  
 109 is awkward, because the meaning of  $*r$  depends on which root location  $r$  was derived from. MVP  
 110 avoids this by rewriting bytecode into a reference-free intermediate form [10]. Shared references are  
 111 inlined:  $\&T$  disappears and reads through it become reads of the underlying value. Mutable references  
 112 are replaced by in/out parameters.

113 **Specification instrumentation.** Before WP runs, the bytecode passes through MVP’s regular  
 114 specification instrumentation. Each function-level requires, aborts\_if, ensures, or modifies  
 115 clause, and each loop invariant, is compiled into an assume or assert at the appropriate program  
 116 point. Calls are instrumented uniformly via the *behavioral predicates* of [13]: requires\_of<f>,  
 117 aborts\_of<f>, ensures\_of<f>, modifies\_of<f>, and result\_of<f> reify a callee’s contract as  
 118 predicates over the call site’s arguments and result. Loops are broken at their headers via the standard  
 119 *cut-point* encoding of Barnett and Leino for unstructured programs [6], with the loop invariant  
 120 playing the role of cut-point predicate: the back-edge is severed, the invariant is asserted before  
 121 and after each iteration, and the loop-modified locations are havoced in between, leaving an acyclic  
 122 fragment through which WP propagates as through any straight-line code.

123 At the input to WP, the bytecode therefore looks like a sequence of pure assignments interleaved with  
 124 assume and assert over plain values; the surface-level features (references, resources, calls, loops,  
 125 user specifications) are gone.

### 126 4 Tactics and Skills

127 The agentic side of the workflow is delivered as a plugin to a general-purpose coding agent: skills  
 128 that instruct it, hooks that check generated Move as it is written, and the Move toolchain exposed as  
 129 tools it can call. The setup is at [2]; we summarize the design.

130 **Inference tactics.** Inference runs in one of three *tactics*, which differ in whether the WP tool is  
 131 available and who decides when to use it.

- 132 – *agent-only*: WP is unavailable. The contract and any loop invariants are derived directly from the  
133 implementation and the contracts of its dependencies, checked as the work proceeds.
  - 134 – *hybrid-guided*: the workflow of Sec. 2, prescribed as a fixed order — run WP over the scope, take  
135 one warned function and write an invariant from the bounded loop-head facts, rerun WP on that  
136 function alone until the warning is gone, simplify what WP derived, then check.
  - 137 – *hybrid-flexible*: WP is available as one inference pass among others, and the agent decides whether  
138 and when to call it and how to combine it with its own reasoning.
- 139 **Skills.** A tactic selects the *plan* — the sequence of tasks and which of them consult WP— and  
140 nothing else. Everything a tactic would otherwise have to rediscover is shared reference material,  
141 identical in all three tactics: the specification language, the rules for writing and editing specs, the  
142 verification sub-workflow, and the toolchain’s capabilities and limits. Only the description of the WP  
143 tool itself is withheld where the tool is absent; the underlying idea of backward reasoning, and what a  
144 loop without an invariant costs, is background every tactic receives.
- 145 Three areas of that guidance carry most of the weight.
- 146 – *Loop invariants*: when to introduce recursive helper functions for iterative quantities, and when  
147 to prefer state-level invariants on resources — assumed by the prover at every call site, including  
148 each loop iteration — over per-loop inductive arguments.
  - 149 – *Simplification*: discard vacuous conditions; replace unbounded quantifiers, a frequent source of  
150 SMT timeouts, with bounded or closed-form equivalents; and require trigger annotations on the  
151 surviving ones.
  - 152 – *Timeout resolution*: a hierarchy of escalations — strengthen state-level invariants, then introduce  
153 spec helpers, then lemmas, then explicit proof scripts — with a strict prohibition on weakening  
154 abort or postconditions to silence a timeout, since this trades a real property for the appearance of  
155 progress.
- 156 Independently of the tactic, every emitted condition is tagged with an [inferred] marker, so that  
157 re-runs are idempotent and never disturb user-written specifications.

## 158 5 Evaluation

- 159 **Setup.** The evaluation setup aims at testing the three different tactics (arms) mentioned above –  
160 *agent-only*, *hybrid-flexible*, and *hybrid-guided*, which are run under the same conditions, using Claude  
161 Code with GLM-5.3 as the underlying model.
- 162 The corpus is drawn from the private Move code of a production perpetual trading system, plus five  
163 targets that are not from it: three functions from a public Move library, and two written for the corpus.  
164 None of the five carries an upstream specification either. A private codebase matters here: the model  
165 cannot have seen these functions, or any specification of them, during training, so a contract has to be  
166 inferred rather than recalled.
- 167 During selection, every candidate target is classified by the features (strata) its contract has to account  
168 for — loops and accumulators, early returns and breaks, arithmetic overflow and underflow, casts and  
169 truncation, mutable references and in-place permutation, function values, quantifiers, composition  
170 through a callee’s contract, and others, 34 such features in all. A target is admitted only if it proves in  
171 a few seconds against a hand-written reference specification. The result is **17 tasks** covering all 34  
172 features across 12 modules, 9 of which contain loops. With 3 arms and 4 replicates this makes **204**  
173 **cells**.
- 174 **Scoring.** A cell reaches *operational success* when its specification compiles, verifies, and passes a  
175 set of tests which check its *completeness*. Completeness is a significant problem for agent-based  
176 inference, since postconditions in particular can be weaker than the actual behavior of the function, and  
177 agents tend to select the shortest path to a solution. The skills make it very clear that completeness is  
178 a requirement, but an agent may ignore this. There is no deductive method for checking completeness  
179 in the general case. WP guarantees completeness only in the presence of complete loop invariants  
180 and complete specs of callees; if completeness is not reached, it may be able to generate a warning.  
181 The agent-only tactic does not have this mechanism.

Table 1: Cost in US dollars per cell by target: mean and median over the four replicates, variation across them (CV), the hybrid mean as a percentage of agent-only’s (% ag.), and whether the specification killed every mutant (✓) or was defeated by one (✗, subscripted with the count where more than one replicate failed). ◦ marks a task-arm whose replicate was lost to an infrastructure fault. Summed over the round the arms cost \$25.54, \$18.25 and \$18.37, i.e.  $0.71\times$  and  $0.72\times$  agent-only.

| target         | agent-only |         |     |                | hybrid-guided |       |         |     |       | hybrid-flexible |       |         |     |       |
|----------------|------------|---------|-----|----------------|---------------|-------|---------|-----|-------|-----------------|-------|---------|-----|-------|
|                | mean \$    | med. \$ | CV% | ok             | mean \$       | % ag. | med. \$ | CV% | ok    | mean \$         | % ag. | med. \$ | CV% | ok    |
| BA-base-012    | 0.380      | 0.393   | 12  | ✓              | 0.251         | 66    | 0.248   | 49  | ✓     | 0.311           | 82    | 0.290   | 28  | ✓     |
| LP-price-021   | 0.197      | 0.191   | 42  | ✓              | 0.228         | 116   | 0.188   | 49  | ✓     | 0.207           | 105   | 0.185   | 30  | ✓     |
| MM-min-013     | 0.126      | 0.098   | 61  | ✓              | 0.141         | 112   | 0.133   | 34  | ✓     | 0.134           | 106   | 0.134   | 10  | ✓     |
| OV-order-006   | 0.097      | 0.091   | 18  | ✓              | 0.076         | 78    | 0.071   | 14  | ✓     | 0.080           | 83    | 0.070   | 37  | ✓     |
| QP-part-025    | 0.698      | 0.660   | 35  | ✗ <sub>2</sub> | 0.490         | 70    | 0.475   | 22  | ✓     | 0.448           | 64    | 0.470   | 42  | ✗     |
| SM-select-022  | 0.406      | 0.375   | 35  | ✓              | 0.434         | 107   | 0.380   | 51  | ✓     | 0.276           | 68    | 0.280   | 34  | ✓     |
| TL-lev-020     | 0.205      | 0.193   | 31  | ✓              | 0.313         | 153   | 0.360   | 34  | ✓     | 0.206           | 100   | 0.195   | 31  | ✓     |
| TR-cancel-026  | 0.442      | 0.403   | 56  | ✓              | 0.335         | 76    | 0.350   | 26  | ✓     | 0.359           | 81    | 0.346   | 17  | ✓     |
| TR-discard-011 | 0.238      | 0.229   | 12  | ✓              | 0.217         | 91    | 0.221   | 22  | ✓     | 0.292           | 122   | 0.275   | 29  | ✓     |
| TR-order-010   | 0.287      | 0.295   | 17  | ✓              | 0.197         | 69    | 0.185   | 29  | ✓     | 0.245           | 85    | 0.261   | 33  | ✓     |
| TS-trial-019   | 0.407      | 0.380   | 56  | ✓              | 0.371         | 91    | 0.285   | 73  | ○     | 0.317           | 78    | 0.272   | 55  | ✓     |
| UC-credits-008 | 0.806      | 0.262   | 146 | ✗              | 0.215         | 27    | 0.197   | 23  | ✓     | 0.389           | 48    | 0.222   | 98  | ✓     |
| VS-contrib-003 | 1.028      | 0.753   | 80  | ✗              | 0.182         | 18    | 0.123   | 68  | ✓     | 0.155           | 15    | 0.159   | 28  | ✓     |
| VS-fees-001    | 0.233      | 0.210   | 34  | ✓              | 0.208         | 90    | 0.224   | 24  | ✓     | 0.200           | 86    | 0.200   | 14  | ✓     |
| VS-redeem-004  | 0.555      | 0.496   | 34  | ✓              | 0.516         | 93    | 0.503   | 16  | ✓     | 0.603           | 109   | 0.496   | 60  | ○     |
| VS-shares-002  | 0.157      | 0.151   | 24  | ✓              | 0.273         | 173   | 0.277   | 49  | ✓     | 0.265           | 169   | 0.241   | 45  | ✓     |
| WU-consume-023 | 0.121      | 0.115   | 27  | ✓              | 0.116         | 96    | 0.109   | 16  | ✓     | 0.106           | 87    | 0.106   | 10  | ✓     |
| all 17         | 0.376      | 0.276   | 110 | 64/68          | 0.268         | 71    | 0.221   | 60  | 67/67 | 0.270           | 72    | 0.233   | 69  | 66/67 |

182 The harness evaluates completeness using *mutation testing* with a set of hand-generated mutants,  
183 changing the verified implementation. To pass the test, each of the mutants needs to be killed — that  
184 is, verification fails when the mutant is verified against the inferred spec.

185 **Cost.** Costs in US dollars use the GLM-5.3 prices recorded for the round: \$1.40 per million uncached  
186 input tokens, \$0.26 per million cached input tokens, and \$4.40 per million output tokens. For the  
187 corresponding token counts  $I_u$ ,  $I_c$ , and  $O$ , the cost is  $cost_{USD} = (1.40I_u + 0.26I_c + 4.40O)/10^6$ .  
188 Cache-creation tokens are currently unchanged and contribute zero; we recompute from the recorded  
189 token counters rather than use the SDK’s cost estimate. Table 1 gives cost per cell, by target. Both  
190 hybrid arms are about 28% cheaper in the mean, and less variable across the round: the coefficient of  
191 variation falls from 110% to 60–69%, and the 12 agent-only cells above \$0.50 become six and five.  
192 The saving is uneven. The hybrids are cheaper on 12 and 11 of the 17 targets, and the round-level  
193 figure is carried by a few of them rather than spread evenly: VS-contrib-003 costs agent-only  
194  $5.6\times$  what it costs hybrid-guided and  $6.6\times$  what it costs hybrid-flexible, while TL-lev-020 and  
195 VS-shares-002 cost  $1.5\times$  and  $1.7\times$  as much under hybrid-guided as without WP. Within a target  
196 the four replicates vary widely in their own right (CV 10–146%), so the per-target ratios are not  
197 individually resolved; the round-level ratio is.

198 The mechanism is visible in the tool counts. The agent-only arm performs about 70% more edits and  
199 three times as many verify calls as hybrid-flexible, and its self-reports describe a different search:  
200 it reaches its contract through counterexamples (52% of answers, against 29% for both hybrids)  
201 and through mutation probes it designed itself (42% against 7–17%). A third of the hybrid answers  
202 instead report WP naming the uninvariant loop, a sentence absent from all 68 agent-only answers.  
203 The arms agree on the product — opaque contracts, recursive spec helpers, loop invariants, at nearly  
204 identical rates — and differ on how the work was found.

205 **Failure points.** Five of the 204 cells failed the mutation test, and 2 more were lost to infras-  
206 tructure faults; Table 1 marks where. Every failure is an agent-only cell but one. Three are  
207 weak\_contract: a mutant survived, in every case leaving normal-result unconstrained. Two  
208 of those are QP-part-025, the in-place permutation, where the surviving class was “right partition  
209 point, wrong contents” — a contract that pins where the partition lands but not what it contains. The  
210 fourth agent-only failure is different in kind: on UC-credits-008 the arm changed the implementa-  
211 tion rather than specify it, which the edit policy forbids. The one hybrid failure is on the same target



212 and of the same class as two of the agent-only ones. The two infrastructure faults are unrelated to  
213 inference (the MCP supervisor restarted the session) and fall one in each hybrid arm.

214 Pooling the two hybrid arms, agent-only left a mutant alive in 4 of its 68 cells against 1 of the hybrids’  
215 134 — 5.9% against 0.75% — and a Fisher exact test gives  $p = 0.045$ , two-sided. The comparison  
216 rests on five events, so it is fragile: the test says the difference is unlikely to be chance, not how large  
217 it is, and a single cell moving either way would change that.

## 218 6 Discussion and Conclusion

### 219 6.1 Interpreting the Results

220 The headline of Sec. 5 is that both WP-backed tactics cost about 28% less than agent-only while  
221 producing mutation-complete specifications in 67/67 hybrid-guided and 66/67 hybrid-flexible cells,  
222 compared with 64/68 agent-only cells; the two cells lost to infrastructure are excluded from these  
223 fractions. The more informative result is how narrow the margin is.

224 **Agent-only does surprisingly well.** Given no WP tool at all, the direct tactic produced a verifying,  
225 mutation-complete specification in 64 of its 68 sessions. It found the same recursive spec helpers and  
226 loop invariants as the hybrids, at nearly the same rates, and it got there by a recognizable method: it  
227 wrote its own mutation probes and let counterexamples drive the repair. The WP tool reduces dollar  
228 cost by about 28% and avoids four failures — real, but not the difference between working and not  
229 working.

230 Two things temper that. The targets are individual functions of a few dozen lines, and while several  
231 need genuine invention — an invented lemma, a permutation contract, a bound that holds only  
232 through a callee’s contract — none requires reasoning across a module or a call graph of any depth.  
233 And the four failures are not spread evenly: they concentrate where the contract has to pin down a  
234 whole data structure rather than a scalar result, which is where a mechanical baseline has the most to  
235 contribute. We expect the gap to widen with the size of the target, and we have no data on that yet.

236 **The next model generation.** The round used a single model, and the obvious question is whether a  
237 stronger one closes the gap. It plausibly narrows it: most of what WP supplies here is a diagnosis  
238 — *this loop has no invariant, and here are its first iterations* — and a model that finds the invariant  
239 unaided needs the diagnosis less. The agent-only sessions that cost the most were the ones searching  
240 without one.

241 We do not read that as WP becoming redundant. What WP derives is exact and free of tokens, and  
242 the hybrid arms also spend their budget elsewhere: on simplifying a machine-shaped contract into a  
243 readable one, which is model work that gets better with the model. A stronger model should improve  
244 both arms, and the interesting measurement is not which wins but whether the mechanical component  
245 keeps removing the same class of failure — the incomplete abort condition, the vacuous postcondition  
246 after a havocked loop — that no amount of fluency prevents.

247 **Open questions.** The evaluation leaves the questions of scale open. Its targets are functions, not  
248 modules; its corpus is one codebase in one domain; and its cost measure is dollars under one model’s  
249 price schedule. We plan to apply inference retrospectively to a public blockchain framework, which  
250 already carries hand-written specifications, and to compare the inferred contracts against them — a  
251 comparison this round cannot make, since its targets deliberately have no specification of their own.

### 252 6.2 Related Work

253 Specification inference predates large language models — Daikon [11] learned likely invariants from  
254 program traces — but the techniques that followed stayed tied to particular property classes and  
255 source languages, and the advent of LLMs has revived interest in the area. Pei et al. [20] showed  
256 that fine-tuned LLMs can predict program invariants statically at quality competitive with dynamic  
257 analysis. Kamath et al. [15] pair an LLM that *proposes* loop invariants with a symbolic checker  
258 that *validates* them, retrying on failure — a propose-and-validate loop that recurs in much of the  
259 subsequent work. For Verus, AlphaVerus [1] bootstraps verified code by self-improving translation  
260 with tree search over verifier feedback, and VeriStruct [21] synthesizes Verus-style specifications and  
261 proofs for Rust functions, modules, and data structures through planner-orchestrated phases. For

smart contracts in Solidity, PropertyGPT [17] retrieves human-written Certora properties as input examples for specification generation. Most directly related to our work, MSG [12] targets the same Move specification language through a skill-based agentic design that consumes verifier feedback beyond a binary success/failure verdict.

Across these systems the LLM remains responsible for the entire specification; the verifier serves as a binary oracle or, in MSG and our setup, as a feedback channel, but never as a contributor of mechanically derived content.

A separate line of work, exemplified by DafnyPro [5] and AutoVerus [22], tackles the complementary problem of synthesizing the auxiliary proof annotations a verifier needs when the specification is already given. Our plugin can also synthesize proof hints (Sec. 2), but this is not the focus of this paper.

Reference elimination, the bytecode transformation underlying our WP analysis (Sec. 3), was introduced in [10], and equivalents exist for Rust. Aeneas [14] compiles Rust to a pure functional model in a target prover (Lean, F\*, Coq, HOL4) via forward and backward functions, while RustHorn [18] encodes a mutable borrow as a pair of its current value and a *prophecy* for its value at the end of the borrow, reducing verification to constrained Horn clauses. Neither infers specifications — in Aeneas the user writes them in the target prover, against the translated model — and both consume the reference-free form only in the forward direction. Our use of the transformation is the opposite: WP runs on the reference-free form, but the conditions it derives have to be carried *back* through the elimination, so that what the user reads and edits is stated over the original references rather than over in/out parameters or prophecy variables. To our knowledge this inversion has not been described before; it is what makes a reference-eliminating pipeline usable for inference rather than only for proving. It is straightforward for *path-based* eliminations such as ours and Aeneas’s, where an eliminated reference still corresponds to a location the source can name. Whether it carries to a prophecy encoding, where the borrow’s final value has no source-level counterpart, we do not know.

Our backend IVL, Boogie [7, 16], also performs WP on the procedural code passed in [6] and passes the result on to the SMT solver. One major difference from our approach is that we aim to generate user-readable conditions, in terms of Move, whereas Boogie’s WP is not intended for external consumption. This is reflected in our mapping of references back to the source language, and in the significant effort put into expression simplification, since raw WP predicates can be very verbose.

In summary, two aspects distinguish our setup from the work above. First, to our knowledge, no prior LLM-based spec-inference system uses a sound mechanical baseline alongside the LLM. WP itself produces a substantial part of the inferred specification — abort conditions, modifies clauses, and consequence-style postconditions — so the AI is left only with what WP cannot recover (loop invariants, high-level conservation laws), and the two streams are validated together by the Move Prover. The AI side runs a propose-observe-refine loop like the systems above; the mechanical baseline narrows the LLM’s search space.

Second, the workflow is delivered as a plugin to a general-purpose, interactive coding agent (Claude Code) via skills, edit-time hooks, and an MCP server, rather than as a custom-built tool: the agent participates in the loop and — most importantly — the user can intervene at any point. To our knowledge, no prior LLM-based inference system for verification languages has this kind of coding-agent integration, including agentic ones such as MSG.

### 6.3 Conclusion

We have presented a specification inference tool for the Move Prover that combines a sound, mechanical WP analysis over Move bytecode with an agentic coding CLI; the prover serves as the oracle for both signals. The two inference streams compose in a single instrumentation pipeline, and the workflow is delivered as a Claude Code plugin — skills, hooks, and an MCP server — so inference runs inside the same conversational loop the user already writes Move in. A controlled comparison on 17 targets drawn from a private perpetual-trading codebase — three tactics, four replicates each — found the WP-backed tactics about 28% cheaper at the stated GLM-5.3 prices and less prone to leaving a contract incomplete, while the unaided agent still produced a complete specification in 64 of its 68 sessions. Whether that margin widens with the size of the target, and whether it survives a stronger model, are the questions we take up next. Neither the WP construction nor the agentic



316 workflow is specific to Move: both rest on reference elimination and on a verifier that consumes  
 317 contracts, so we expect the approach to transfer to Rust-like languages and their verifiers.

## 318 References

- 319 [1] Pranjal Aggarwal, Bryan Parno, and Sean Welleck. AlphaVerus: Bootstrapping formally verified  
 320 code generation through self-improving translation and tree refinement. In *Proceedings of the*  
 321 *42nd International Conference on Machine Learning (ICML 2025)*, volume 267 of *Proceedings*  
 322 *of Machine Learning Research*, 2025.
- 323 [2] Anonymous. Specification-inference plugin for the move prover, 2026. Source available on  
 324 request; released with the camera-ready.
- 325 [3] Aptos Labs. The Aptos Framework Book. <https://aptos-labs.github.io/framework-book/>, 2023.  
 326 Accessed: 2026-05-09.
- 327 [4] Aptos Labs. The Move on Aptos Book. <https://aptos-labs.github.io/move-book/>, 2023.  
 328 Accessed: 2026-05-09.
- 329 [5] Debangshu Banerjee, Olivier Bouissou, and Stefan Zetsche. DafnyPro: LLM-assisted auto-  
 330 mated verification for Dafny programs, 2026. URL <https://arxiv.org/abs/2601.05385>. Dafny  
 331 Workshop, POPL 2026.
- 332 [6] Mike Barnett and K. Rustan M. Leino. Weakest-precondition of unstructured programs. In  
 333 *Proceedings of the 6th ACM SIGPLAN-SIGSOFT Workshop on Program Analysis for Software*  
 334 *Tools and Engineering*, pages 82–87, New York, NY, USA, 2005. Association for Computing  
 335 Machinery. ISBN 1595932399. doi: 10.1145/1108792.1108813.
- 336 [7] Mike Barnett, Bor-Yuh Evan Chang, Robert DeLine, Bart Jacobs, and K Rustan M Leino.  
 337 Boogie: A modular reusable verifier for object-oriented programs. In *International Symposium*  
 338 *on Formal Methods for Components and Objects*, pages 364–387. Springer, 2005.
- 339 [8] Patrick Cousot and Nicolas Halbwachs. Automatic discovery of linear restraints among variables  
 340 of a program. In *Proceedings of the 5th ACM SIGACT-SIGPLAN Symposium on Principles of*  
 341 *Programming Languages (POPL '78)*, pages 84–96. ACM, 1978. doi: 10.1145/512760.512770.
- 342 [9] Edsger W. Dijkstra. Guarded commands, nondeterminacy and formal derivation of programs.  
 343 *Communications of the ACM*, 18(8):453–457, 1975. doi: 10.1145/360933.360975.
- 344 [10] David L. Dill, Wolfgang Grieskamp, Junkil Park, Shaz Qadeer, Meng Xu, and Jingyi Emma  
 345 Zhong. Fast and reliable formal verification of smart contracts with the move prover. In *Tools*  
 346 *and Algorithms for the Construction and Analysis of Systems (TACAS 2022)*, volume 13243 of  
 347 *Lecture Notes in Computer Science*, pages 183–200. Springer, 2022. doi: 10.1007/978-3-030-9  
 348 9524-9\_10.
- 349 [11] Michael D. Ernst, Jake Cockrell, William G. Griswold, and David Notkin. Dynamically  
 350 discovering likely program invariants to support program evolution. *IEEE Transactions on*  
 351 *Software Engineering*, 27(2):99–123, 2001. doi: 10.1109/32.908957.
- 352 [12] Yu-Fu Fu, Meng Xu, and Taesoo Kim. Agentic specification generator for Move programs. In  
 353 *Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineer-*  
 354 *ing (ASE 2025)*, pages 1286–1298, 2025. doi: 10.1109/ASE63991.2025.00110.
- 355 [13] Wolfgang Grieskamp, Teng Zhang, Vineeth Kashyap, and Jake Silverman. Formal verification  
 356 of imperative first-class functions in Move. *CoRR*, abs/2605.10007, 2026. URL <https://arxiv.org/abs/2605.10007>. Preprint; to appear in FMCAD 2026.
- 358 [14] Son Ho and Jonathan Protzenko. Aeneas: Rust verification by functional translation. *Proc.*  
 359 *ACM Program. Lang.*, 6(ICFP):116:1–116:31, 2022. doi: 10.1145/3547647.
- 360 [15] Adharsh Kamath, Aditya Senthilnathan, Saikat Chakraborty, Pantazis Deligiannis, Shuvendu K.  
 361 Lahiri, Akash Lal, Aseem Rastogi, Subhajit Roy, and Rahul Sharma. Finding inductive loop  
 362 invariants using large language models, 2023. URL <https://arxiv.org/abs/2311.07948>.

- 363 [16] K. Rustan M. Leino and Philipp Rümmer. A polymorphic intermediate verification language:  
364 Design and logical encoding. In Javier Esparza and Rupak Majumdar, editors, *TACAS*, pages  
365 312–327, Berlin, Heidelberg, 2010. Springer Berlin Heidelberg. ISBN 978-3-642-12002-2.
- 366 [17] Ye Liu, Yue Xue, Daoyuan Wu, Yuqiang Sun, Yi Li, Miaolei Shi, and Yang Liu. PropertyGPT:  
367 LLM-driven formal verification of smart contracts through retrieval-augmented property gen-  
368 eration. In *Network and Distributed System Security Symposium (NDSS 2025)*, 2025. doi:  
369 10.14722/ndss.2025.241357.
- 370 [18] Yusuke Matsushita, Takeshi Tsukada, and Naoki Kobayashi. RustHorn: CHC-based verification  
371 for Rust programs. *ACM Trans. Program. Lang. Syst.*, 43(4):1–54, 2021. doi: 10.1145/3462205.
- 372 [19] Junkil Park, Teng Zhang, Wolfgang Grieskamp, Meng Xu, Gerardo Di Giacomo, Kundu  
373 Chen, Yi Lu, and Robert Chen. Securing Aptos Framework with Formal Verification. In *5th*  
374 *International Workshop on Formal Methods for Blockchains (FMBC 2024)*, volume 118 of *Open*  
375 *Access Series in Informatics (OASIs)*, pages 9:1–9:16. Schloss Dagstuhl – Leibniz-Zentrum für  
376 Informatik, 2024. doi: 10.4230/OASIs.FMBC.2024.9.
- 377 [20] Kexin Pei, David Bieber, Kensen Shi, Charles Sutton, and Pengcheng Yin. Can large language  
378 models reason about program invariants? In *Proceedings of the 40th International Conference*  
379 *on Machine Learning (ICML 2023)*, volume 202 of *Proceedings of Machine Learning Research*,  
380 pages 27496–27520. PMLR, 2023. URL <https://proceedings.mlr.press/v202/pei23a.html>.
- 381 [21] Chuyue Sun, Yican Sun, Daneshvar Amrollahi, Ethan Zhang, Shuvendu K. Lahiri, Shan Lu,  
382 David L. Dill, and Clark Barrett. VeriStruct: AI-assisted automated verification of data-  
383 structure modules in Verus. In *Tools and Algorithms for the Construction and Analysis of*  
384 *Systems (TACAS 2026)*, Lecture Notes in Computer Science. Springer, 2026. URL <https://arxiv.org/abs/2510.25015>. arXiv:2510.25015.
- 386 [22] Chenyuan Yang, Xuheng Li, Md Rakib Hossain Misu, Jianan Yao, Weidong Cui, Yeyun Gong,  
387 Chris Hawblitzel, Shuvendu K. Lahiri, Jacob R. Lorch, Shuai Lu, Fan Yang, Ziqiao Zhou, and  
388 Shan Lu. AutoVerus: Automated proof generation for Rust code. *Proc. ACM Program. Lang.*,  
389 9(OOPSLA2), 2025. doi: 10.1145/3763174.
- 390 [23] Z.ai. GLM-5.3. <https://huggingface.co/zai-org/GLM-5.3>, 2026. Model card. Accessed:  
391 2026-09-04.